

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Single Correct

The number of elements in the set

$\{x \in \mathbb{R} : (|x| - 3)|x + 4| = 6\}$ is equal to

(1) 3

(2) 2

(3) 4

(4) 1

Q2: 16 March (Shift 2) - Single Correct

Let $P(x) = x^2 + bx + c$ be a quadratic polynomial with real coefficients such that $\int_0^1 P(x) dx = 1$ and $P(x)$ leaves remainder 5 when it is divided by $(x - 2)$. Then the value of $9(b + c)$ is equal to:

(1) 9

(2) 15

(3) 7

(4) 11

Q3: 17 March (Shift 1) - Single Correct

The value of $4 + \frac{1}{5 + \frac{1}{4 + \frac{1}{5 + \frac{1}{4 + \dots \infty}}}}$

(1) $2 + \frac{2}{5}\sqrt{30}$

(2) $2 + \frac{4}{\sqrt{5}}\sqrt{30}$

(3) $4 + \frac{4}{\sqrt{5}}\sqrt{30}$

(4) $5 + \frac{2}{5}\sqrt{30}$

Q4: 18 March (Shift 1) - Single Correct

The value of $3 + \frac{1}{4 + \frac{1}{3 + \frac{1}{4 + \frac{1}{3 + \dots \infty}}}}$ is equal to

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(1) $1.5 + \sqrt{3}$

(2) $2 + \sqrt{3}$

(3) $3 + 2\sqrt{3}$

(4) $4 + \sqrt{3}$

Q5: 18 March (Shift 2) - Numerical

If $f(x)$ and $g(x)$ are two polynomials such that the polynomial $P(x) = f(x^3) + xg(x^3)$ is divisible by

$x^2 + x + 1$, then $P(1)$ is equal to _____

Answer Key

Q1 (2)

Q2 (3)

Q3 (1)

Q4 (1)

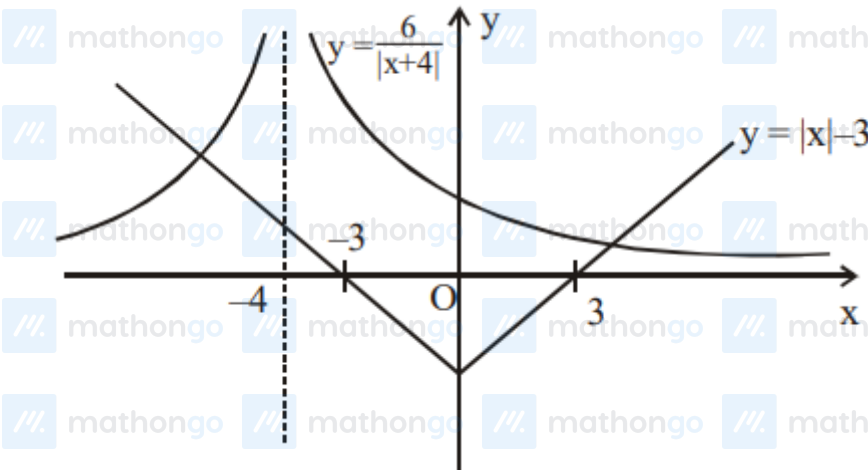
Q5 (0)

Q1 (2)

$$x \neq -4$$

$$(|x| - 3)(|x + 4|) = 6$$

$$\Rightarrow |x| - 3 = \frac{6}{|x+4|}$$



No. of solutions = 2

Q2 (3)

$$\int_0^1 (x^2 + bx + c) dx = 1$$

$$\frac{1}{3} + \frac{b}{2} + c = 1 \Rightarrow \frac{b}{2} + c = \frac{2}{3}$$

$$3b + 6c = 4$$

$$P(2) = 5$$

$$4 + 2b + c = 5$$

$$2b + c = 1$$

From (1)&(2)

$$b = \frac{2}{9} \quad \&c = \frac{5}{9}$$

$$9(b + c) = 7$$

Q3 (1)

Hints and Solutions

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$$y = 4 + \frac{1}{\left(5 + \frac{1}{y}\right)}$$

$$y - 4 = \frac{y}{(5y+1)}$$

$$5y^2 - 20y - 4 = 0$$

$$y = \frac{20 \pm \sqrt{480}}{10}$$

$$y = \frac{20 - \sqrt{480}}{10} \rightarrow \text{rejected}$$

$$y = 2 + \sqrt{\frac{480}{100}}$$

Correct with Option (A)

Q4 (1)

$$\text{Let } x = 3 + \frac{1}{4 + \frac{1}{3 + \frac{1}{4 + \frac{1}{3 + \dots \infty}}}}$$

$$\text{So, } x = 3 + \frac{1}{4 + \frac{1}{x}} = 3 + \frac{1}{\frac{4x+1}{x}}$$

$$\Rightarrow (x - 3) = \frac{x}{(4x+1)}$$

$$\Rightarrow (4x + 1)(x - 3) = x$$

$$\Rightarrow 4x^2 - 12x + x - 3 = x$$

$$\Rightarrow 4x^2 - 12x - 3 = 0$$

$$x = \frac{12 \pm \sqrt{(12)^2 + 12 \times 4}}{2 \times 4} = \frac{12 \pm \sqrt{12(16)}}{8}$$

$$= \frac{12 \pm 4 \times 2\sqrt{3}}{8} = \frac{3 \pm 2\sqrt{3}}{2}$$

$$x = \frac{3}{2} \pm \sqrt{3} = 1.5 \pm \sqrt{3}$$

But only positive value is accepted

$$\text{So, } x = 1.5 + \sqrt{3}$$

Q5 (0)

$$P(x) = f(x^3) + xg(x^3)$$

$$P(1) = f(1) + g(1) \dots (1)$$

Now P(x) is divisible by $x^2 + x + 1$

$$\Rightarrow P(x) = Q(x)(x^2 + x + 1)$$

$P(w) = 0 = P(w^2)$ where w, w^2 are non-real cube

Hints and Solutions

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roots of units

$$P(x) = f(x^3) + xg(x^3)$$

$$P(w) = f(w^3) + wg(w^3) = 0$$

$$f(1) + wg(1) = 2 \dots (2)$$

$$P(w^2) = f(w^6) + w^2g(w^6) = 0$$

$$f(1) + w^2g(1) = 0 \dots (3)$$

$$(2) + (3)$$

$$\Rightarrow 2f(1) + (w + w^2)g(1) = 0$$

$$2f(1) = g(1) \dots (4)$$

$$(2) - (3)$$

$$\Rightarrow (w - w^2)g(1) = 0$$

$$g(1) = 0 = f(1) \text{ from (4)}$$

$$\text{from (1) } P(1) = f(1) + g(1) = 0$$